Orders, Fermat and Euler

L4-NT1 Level 4 Olympiad · Number theory

Kevin Hu

November 2026

How the order of a number modulo n controls divisibility of powers, with Fermat, Euler, primitive roots and lifting the exponent.
By the end you can find and use the order of a number modulo n, and prove that this order divides $\varphi(n)$; pin down the prime factors of expressions such as $a^m + 1$ using orders; find the exact power of a prime dividing $a^n - b^n$ with lifting the exponent.
Prerequisites: Modular arithmetic (L3-NT1), Integer equations (L3-NT2). **Contests:** BMO1, MOG, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

Does 7 ever divide $2^n + 1$? Write out the powers of 2 modulo 7. They run $2, 4, 1, 2, 4, 1, \ldots$, repeating every three steps, and $-1 \equiv 6$ never turns up. So $7 \nmid 2^n + 1$ for every n. Now consider $3^n + 1$. The powers of 3 modulo 7 are $3, 2, 6, 4, 5, 1$ and then repeat, and $3^3 \equiv 6 \equiv -1$, so $7 \mid 3^n + 1$ exactly when $n \equiv 3 \pmod 6$.

In general the powers of a modulo n are periodic, and the length of the period determines which exponents satisfy a given congruence. This unit studies that period.

1.1 The order of a number

Let $n \geq 2$ and let a be an integer with $\gcd(a, n) = 1$. The **order** of a modulo n, written $\operatorname{ord}_n(a)$, is the smallest positive integer d with $a^d \equiv 1 \pmod n$.

Such a d exists. There are only n residues, so two of $a^0, a^1, \ldots, a^n$ agree, say $a^i \equiv a^j$ with $i < j$. Because a is coprime to n we may cancel a i times, which leaves $a^{j-i} \equiv 1$. So some positive power is 1, and hence there is a smallest one.

Order Lemma.
Let $d = \operatorname{ord}_n(a)$. Then $a^k \equiv 1 \pmod n$ if and only if $d \mid k$. Consequently $a^i \equiv a^j \pmod n$ if and only if $i \equiv j \pmod d$.

Proof. If $k = dm$ then $a^k = (a^d)^m \equiv 1$. Conversely, suppose $a^k \equiv 1$ and divide to get $k = qd + r$ with $0 \leq r < d$. Then $1 \equiv a^k = (a^d)^q a^r \equiv a^r$. Since $r < d$ and d is the smallest positive exponent giving 1, we must have $r = 0$. The second statement follows by cancelling the smaller power. □

Two consequences of the lemma are used constantly. First, if $a^m \equiv 1$ and $a^k \equiv 1$ then d divides both, so $a^{\gcd(m,k)} \equiv 1$. Second, to show that the order is exactly d, we only need to check that $a^d \equiv 1$ and that $a^{d/q} \not\equiv 1$ for every prime $q \mid d$, because every proper divisor of d divides some d/q.

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Example 1.
Find the smallest positive integer n such that 1001 divides $2^n - 1$.

Solution.
One could multiply by 2 modulo 1001 until 1 appears, but splitting the modulus is quicker. We have $1001 = 7 \cdot 11 \cdot 13$, and $1001 \mid 2^n - 1$ if and only if each of 7, 11, 13 divides $2^n - 1$. By the Order Lemma that happens if and only if n is a multiple of each of the three orders, so the answer is their lowest common multiple.
Modulo 7, $2^3 = 8 \equiv 1$ and $2 \not\equiv 1$, so the order is 3. Modulo 11, $2^5 = 32 \equiv -1$, so $2^{10} \equiv 1$ while $2^5 \not\equiv 1$ and $2^2 = 4 \not\equiv 1$. The primes dividing 10 are 2 and 5, so by the test above the order is 10. Modulo 13, $2^6 = 64 \equiv -1$, so $2^{12} \equiv 1$, while $2^6 \not\equiv 1$ and $2^4 = 16 \equiv 3 \not\equiv 1$. The primes dividing 12 are 2 and 3, so the order is 12.
Hence the answer is $\operatorname{lcm}(3, 10, 12) = 60$.

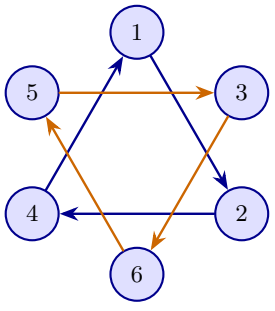
1.2 Fermat and Euler

Write $\varphi(n)$ for the number of integers in $\{1, 2, \ldots, n\}$ that are coprime to n. For a prime, $\varphi(p) = p - 1$.

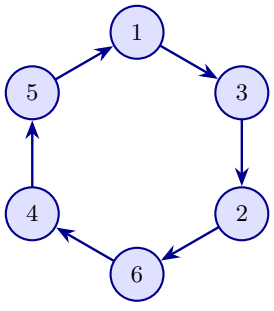
Euler's Theorem.
If $\gcd(a, n) = 1$ then $a^{\varphi(n)} \equiv 1 \pmod{n}$. In particular (Fermat's Little Theorem) $a^{p-1} \equiv 1 \pmod{p}$ for a prime p not dividing a. Combined with the Order Lemma, this gives $\operatorname{ord}_n(a) \mid \varphi(n)$, and $\operatorname{ord}_p(a) \mid p - 1$.

Proof. Let $r_1, \ldots, r_k$ be the residues coprime to n, where $k = \varphi(n)$. The numbers $ar_1, \ldots, ar_k$ are also coprime to n, and no two are congruent, since $ar_i \equiv ar_j$ gives $r_i \equiv r_j$ after cancelling a. So they are the same k residues in a different order, and multiplying everything together gives $a^k r_1 \cdots r_k \equiv r_1 \cdots r_k$. The product $r_1 \cdots r_k$ is coprime to n, so the factor $r_1 \cdots r_k$ cancels. □

For a prime, a second proof explains why the order divides $p - 1$. Multiplication by a permutes the nonzero residues, and the permutation splits into cycles. The cycle through r is $r, ar, a^2 r, \ldots$, and this cycle first returns to r after exactly $\operatorname{ord}_p(a)$ steps, since $a^k r \equiv r$ means $a^k \equiv 1$. So every cycle has length $\operatorname{ord}_p(a)$, and since the cycles partition the $p - 1$ nonzero residues, $\operatorname{ord}_p(a) \mid p - 1$.



$x \mapsto 2x \pmod 7$
two cycles of length 3



$x \mapsto 3x \pmod 7$
one cycle of length 6

Orders most often enter olympiad problems through expressions like $a^m + 1$. If an odd prime p divides $a^m + 1$, then $a^{2m} \equiv 1$ but $a^m \equiv -1 \not\equiv 1$, so the order divides $2m$ but does not divide m.

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Prime factors of $a^{2^k}+1$.
If an odd prime p divides $a^{2^k}+1$, then $\operatorname{ord}_p(a) = 2^{k+1}$, and therefore $p \equiv 1 \pmod{2^{k+1}}$.

Proof. The order divides 2^{k+1}, so the order is 2^j for some $j \le k+1$. If $j \le k$ then $a^{2^k} = (a^{2^j})^{2^{k-j}} \equiv 1$, contradicting $a^{2^k} \equiv -1$ (and $1 \not\equiv -1$ because p is odd). So $j = k+1$, and $2^{k+1} \mid p-1$. □

Hence every odd prime factor of n^2+1 is 1 modulo 4, and every odd prime factor of n^4+1 is 1 modulo 8.

Example 2.
Find all pairs of primes (p, q) such that $p \mid 2^q+1$ and $q \mid 2^p+1$.

Solution.
Small cases suggest that $p = q = 3$ is the only answer. To relate p and q we use the order of 2 modulo each prime.
Both 2^q+1 and 2^p+1 are odd, so p and q are odd. From $p \mid 2^q+1$ we get $2^{2q} \equiv 1 \pmod{p}$ while $2^q \equiv -1 \not\equiv 1$. Let $d = \operatorname{ord}_p(2)$. Then $d \mid 2q$ and $d \nmid q$. The divisors of $2q$ are $1, 2, q, 2q$, and the condition $d \nmid q$ removes 1 and q. If $d = 2$ then $p \mid 2^2 - 1 = 3$, so $p = 3$. If $d = 2q$ then $2q \mid p-1$ by Fermat, so $p > q$.
So either $p = 3$ or $p > q$. By symmetry, either $q = 3$ or $q > p$. We cannot have both $p > q$ and $q > p$, so at least one of the primes is 3. If $p = 3$ then $q \mid 2^3+1 = 9$, so $q = 3$. If $q = 3$ then likewise $p = 3$. The only pair is $(3, 3)$.

Example 3.
Let n be an integer. Prove that every prime factor p of $n^4 - n^2 + 1$ satisfies $p \equiv 1 \pmod{12}$.

Solution.
The expression is not of the form a^m+1, so the lemma above does not apply directly. Instead we use the factorisation $(n^2+1)(n^4-n^2+1) = n^6+1$. So $p \mid n^6+1$, giving $n^6 \equiv -1$ and $n^{12} \equiv 1 \pmod{p}$.
The number $n^4 - n^2 + 1 = n^2(n^2-1)+1$ is odd, so p is odd. Also $p \nmid n$, otherwise $n^4 - n^2 + 1 \equiv 1$. Let $d = \operatorname{ord}_p(n)$. Then $d \mid 12$ and $d \nmid 6$, which leaves $d = 4$ or $d = 12$.
The case $d = 4$ has to be ruled out separately, and many solutions skip this case. Suppose $d = 4$. Then $n^4 \equiv 1$, so $0 \equiv n^4 - n^2 + 1 \equiv 2 - n^2$, giving $n^2 \equiv 2$ and so $n^4 \equiv 4$. Together with $n^4 \equiv 1$ this forces $p \mid 3$. But 3 never divides $n^4 - n^2 + 1$. If $3 \mid n$ the value is 1 modulo 3, and otherwise $n^2 \equiv 1$ and the value is $1 - 1 + 1 = 1$ modulo 3. So $d = 4$ is impossible, $d = 12$, and $12 \mid p-1$. □

1.3 Primitive roots, and why $x^d \equiv 1$ has few solutions

Lagrange's Theorem.
Let p be prime and f a polynomial with integer coefficients and degree $d \ge 1$, whose leading coefficient is not divisible by p. Then $f(x) \equiv 0 \pmod{p}$ has at most d solutions modulo p.

Proof. Induct on d. The case $d = 1$ is clear because the leading coefficient is invertible. If r is a root, polynomial division gives $f(x) = (x-r)g(x) + f(r)$ with g of degree $d-1$, integer coefficients and the same leading coefficient. If $s \not\equiv r$ is another root, then $(s-r)g(s) \equiv 0$, and since p is prime and $p \nmid s-r$, we get $g(s) \equiv 0$. So the other roots are roots of g, and there are at most $d-1$

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of them. □

Note that the modulus must be prime, since $x^2 \equiv 1 \pmod 8$ has four solutions. A consequence is that k-th powers take few values modulo p (see L4-NT2).

A harder fact, which we state without proof, is the **Primitive Root Theorem**. For every prime p there is a residue g with $\operatorname{ord}_p(g) = p-1$. Then $g, g^2, \ldots, g^{p-1}$ run through all the nonzero residues, and for every $d \mid p-1$ the residue $g^{(p-1)/d}$ has order exactly d. So *an element of order d exists modulo p if and only if $d \mid p-1$*. The proof counts elements of each order using Lagrange's Theorem. The argument is not hard, and is left as an exercise for the reader.

1.4 Lifting the exponent

Orders tell you when p divides $a^n - b^n$. Lifting the exponent gives the exact power of p that divides this difference. Write $v_p(m)$ for the exponent of p in m.

Lifting the Exponent (LTE).
Let p be an odd prime and let a, b be integers with $p \mid a-b$ and $p \nmid ab$. Then for every positive integer n,

$$v_p(a^n - b^n) = v_p(a-b) + v_p(n).$$

For $p = 2$, if a, b are odd and n is even, then $v_2(a^n - b^n) = v_2(a-b) + v_2(a+b) + v_2(n) - 1$.

Proof for odd p. Step 1. If $p \nmid n$, write $a^n - b^n = (a-b)S$ with $S = a^{n-1} + a^{n-2}b + \cdots + b^{n-1}$. As $a \equiv b$, each of the n terms of S is congruent to a^{n-1}, so $S \equiv na^{n-1} \not\equiv 0 \pmod p$, and $v_p(a^n - b^n) = v_p(a-b)$.

Step 2 is the case $n = p$. Put $a = b + t$ with $p \mid t$. By the binomial theorem $(b+t)^m \equiv b^m + mtb^{m-1} \pmod{t^2}$, and $p^2 \mid t^2$, so modulo p^2

$$S = \sum_{i=0}^{p-1} (b+t)^{p-1-i} b^i \equiv pb^{p-1} + tb^{p-2} \sum_{i=0}^{p-1} (p-1-i) = pb^{p-1} + tb^{p-2} \cdot p \cdot \frac{p-1}{2}.$$

Because p is odd, $\frac{p-1}{2}$ is an integer and the last term is divisible by pt, hence by p^2. So $S \equiv pb^{p-1} \pmod{p^2}$, giving $v_p(S) = 1$ and $v_p(a^p - b^p) = v_p(a-b) + 1$.

For general $n = p^k m$ with $p \nmid m$, apply Step 2 to a, b, then to a^p, b^p, and so on k times (the hypotheses survive each time), and finish with Step 1. □

For $p = 2$, write $n = 2^k m$ with m odd. The odd part changes nothing, by the argument of Step 1 (a sum of m odd terms is odd), and $a^{2^k} - b^{2^k} = (a-b)(a+b)(a^2+b^2)(a^4+b^4)\cdots(a^{2^{k-1}} + b^{2^{k-1}})$, where each factor $a^{2^j} + b^{2^j}$ with $j \geq 1$ is a sum of two odd squares, so this factor is 2 modulo 4. That gives the extra $k-1$.

Example 4.
Prove that for every positive integer k, the order of 2 modulo 3^k is $2 \cdot 3^{k-1}$.

Solution.
We need to know which n make $3^k \mid 2^n - 1$. LTE counts powers of 3, but LTE needs $3 \mid a-b$, and $3 \nmid 2-1$. So we first find the order of 2 modulo 3. Since $2 \equiv -1$, we have $2^n \equiv 1 \pmod 3$ if and only if n is even.
So if $3^k \mid 2^n - 1$ then $n = 2m$, and $2^n - 1 = 4^m - 1$ where now $3 \mid 4-1$. LTE gives

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$v_3(4^m - 1) = v_3(3) + v_3(m) = 1 + v_3(n)$. Therefore $3^k \mid 2^n - 1$ if and only if n is even and $v_3(n) \geq k - 1$, that is, if and only if $2 \cdot 3^{k-1} \mid n$. The smallest such positive n is $2 \cdot 3^{k-1}$, which is the order. □

So 2 is a primitive root modulo every power of 3. The pattern "find the order modulo p, then lift with LTE" is the standard method for AIME questions asking for the least n with $p^k \mid a^n - b^n$.

Common traps.

- The order only exists when $\gcd(a, n) = 1$. Check this before you write $\mathrm{ord}_n(a)$.
- The order divides $p - 1$ but is usually smaller. Never assume 2 has order $p - 1$ without checking.
- From $a^m \equiv -1$ you learn that the order divides $2m$ but not m. List all such divisors, including the small ones. The cases $d = 2$ in Example 2 and $d = 4$ in Example 3 are where marks are most often lost.
- LTE needs $p \mid a - b$ and $p \nmid ab$. If $p \nmid a - b$, first raise to the order. Using the odd-prime formula at $p = 2$ is wrong, since $v_2(3^2 - 1) = 3$, not $v_2(2) + v_2(2) = 2$.
- Lagrange's bound needs a prime modulus.

1.5 How to spot this in a contest

Think orders whenever a problem involves $a^n \pm 1$, $a^n - b^n$ or "for which n does p divide". The answer is usually a congruence class modulo an order. Prime factors of $n^2 + 1$, $n^4 + 1$ or anything dividing $n^k \pm 1$ point to the "order divides $2m$ but not m" argument. "Largest power of p dividing" or "least n with $p^k \mid \ldots$" is LTE, after first reaching the order modulo p. Exponent towers are reduced by working modulo the order (or φ) in the exponent.

Finally, a standard opening (the smallest prime factor trick) for problems of the form $n \mid a^n - b^n$ or $n \mid a^n + 1$. Look at the **smallest prime factor** p of n. The order of a (or of ab^{-1}) modulo p divides both something built from n and $p - 1$, and since every prime factor of n is at least p while every prime factor of $p - 1$ is less than p, $\gcd(n, p - 1) = 1$, so the order is usually forced to divide 2. Practice problem 6 uses this argument.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ Find the order of 13 modulo 100, and use this order to find the last two digits of 13^{2026}.

Problem 2 ⋆⋆ Find all positive integers n such that 13 divides $2^n + 3^n$.

Problem 3 ⋆⋆ Find the largest integer k such that 10^k divides $3^{2000} - 1$.

Problem 4 ⋆⋆⋆ Find the sum of all primes $p < 200$ for which there is an integer n with $p \mid$

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$n^6 + n^3 + 1$. (You may use the Primitive Root Theorem.)

Problem 5 ⋆⋆⋆ Prove that there are infinitely many primes of the form $8k + 1$.

Problem 6 ⋆⋆⋆⋆ Let n be an odd positive integer such that n divides $3^n + 1$. Prove that $n = 1$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT BMO1 2005/06 Problem 1** ⋆
   Split 720 into prime powers, and Fermat's little theorem handles the factor of 5.
2. **AIME 1983 Problem 6** ⋆
   Euler's theorem modulo 49, or expand around 7.
3. **UKMT BMO1 2006/07 Problem 1** ⋆⋆
   Factorise as a difference of squares as far as the factorisation goes, then ask which primes can divide each factor.
4. **AIME I 2011 Problem 11** ⋆⋆
   Split 1000 as 8 times 125 and find the order of 2 modulo 125.
5. **UKMT BMO1 2013/14 Problem 3** ⋆⋆⋆
   Write the number using a power of 10, then lift the exponent.
6. **AIME I 2020 Problem 12** ⋆⋆⋆
   Treat each prime power separately. Reach the order first, then lift the exponent.
7. **AIME I 2019 Problem 14** ⋆⋆⋆
   What does p dividing $a^8 + 1$ force about the order of a modulo p?
8. **AIME II 2021 Problem 13** ⋆⋆⋆
   Work modulo 8 and 125 separately and find the period of each part.
9. **UKMT BMO1 2024/25 Problem 5** ⋆⋆⋆⋆
   The residues x with x^6 congruent to 1 form a small, structured set. Work out what this set looks like.
10. **AIME I 2024 Problem 13** ⋆⋆⋆⋆
    Use the order of n to restrict p, then lift a root from p to p^2.
11. **AIME II 2023 Problem 15** ⋆⋆⋆⋆
    The order of 2 modulo 23 controls how the multiples change as n grows.

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4 Hints

1. Work modulo 4 and modulo 25 separately. The order modulo 100 is the lowest common multiple of the two orders.
2. Divide through by 3^n. The condition is about a single number raised to the power n.
3. Find the powers of 2 and of 5 separately. For 5, you need to reach the order of 3 modulo 5 before LTE applies.
4. Multiply by $n^3 - 1$. What can the order of n modulo p be?
5. Suppose there are only finitely many and build a number of the form $N^4 + 1$ from them. What do its prime factors look like?
6. Suppose $n > 1$ and let p be the smallest prime factor of n. The order of 3 modulo p divides two numbers with very little in common.

5 Solutions

Problem 1

Find the order of 13 *modulo* 100*, and use this order to find the last two digits of* 13^{2026}.

Modulo 4 we have $13 \equiv 1$, so the order there is 1. Modulo 25 the order divides $\varphi(25) = 20$. Now $13^2 = 169 \equiv 19 \equiv -6$, so $13^4 \equiv 36 \equiv 11$ and $13^5 \equiv 143 \equiv -7$, giving $13^{10} \equiv 49 \equiv -1 \pmod{25}$. Hence $13^{20} \equiv 1$, while $13^{10} \not\equiv 1$ and $13^4 \not\equiv 1$. Every proper divisor of 20 divides 10 or 4, so the order modulo 25 is 20.

A power 13^n is 1 modulo 100 exactly when the power is 1 modulo both 4 and 25, that is, when $20 \mid n$. So $\operatorname{ord}_{100}(13) = 20$. Since $2026 = 20 \cdot 101 + 6$, the Order Lemma gives $13^{2026} \equiv 13^6 \pmod{100}$. From $13^2 \equiv 69$ we get $13^4 \equiv 69^2 = 4761 \equiv 61$ and $13^6 \equiv 61 \cdot 69 = 4209 \equiv 9$. The last two digits are 09.

Answer: The order is 20; the last two digits are 09.

Problem 2

Find all positive integers n *such that* 13 *divides* $2^n + 3^n$.

Since $3 \cdot 9 = 27 \equiv 1 \pmod{13}$, the inverse of 3 is 9. Multiplying by 9^n, the condition $2^n + 3^n \equiv 0$ becomes $(2 \cdot 9)^n \equiv -1$, that is, $5^n \equiv -1 \pmod{13}$.

Note that $5^2 = 25 \equiv -1$, so $5^4 \equiv 1$ and the order of 5 is 4. The powers of 5 cycle through $5, -1, -5, 1$, so $5^n \equiv -1$ exactly when $n \equiv 2 \pmod{4}$. These are the required n. For example $2^2 + 3^2 = 13$.

Answer: Exactly the n with $n \equiv 2 \pmod{4}$.

Problem 3

Find the largest integer k *such that* 10^k *divides* $3^{2000} - 1$.

For the prime 2, the numbers 3 and 1 are odd and 2000 is even, so the $p = 2$ form of LTE gives $v_2(3^{2000} - 1) = v_2(3 - 1) + v_2(3 + 1) + v_2(2000) - 1 = 1 + 2 + 4 - 1 = 6$.

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For the prime 5, LTE cannot be applied to $3^{2000} - 1^{2000}$ directly, because $5 \nmid 3 - 1$. But $3^2 = 9 \equiv -1 \pmod 5$, so $\text{ord}_5(3) = 4$, and $3^{2000} - 1 = 81^{500} - 1$ with $5 \mid 81 - 1$. LTE now gives $v_5(81^{500} - 1) = v_5(80) + v_5(500) = 1 + 3 = 4$.

So $3^{2000} - 1$ is divisible by 2^6 and by 5^4 but not by 5^5, and the largest power of 10 dividing this number is 10^4. Thus $k = 4$.

Answer: 4

Problem 4

Find the sum of all primes $p < 200$ for which there is an integer n with $p \mid n^6 + n^3 + 1$. (You may use the Primitive Root Theorem.)

Note first that $n^6 + n^3 + 1$ is always odd, since $n^6 + n^3 = n^3(n^3 + 1)$ is even, so $p \neq 2$. Also $(n^3 - 1)(n^6 + n^3 + 1) = n^9 - 1$.

Suppose $p \mid n^6 + n^3 + 1$. Then $p \nmid n$ and $n^9 \equiv 1 \pmod p$, so $d = \text{ord}_p(n)$ divides 9. If $d \mid 3$ then $n^3 \equiv 1$, so $n^6 + n^3 + 1 \equiv 3$ and $p = 3$. Otherwise $d = 9$, and $9 \mid p - 1$ by Fermat's Little Theorem.

Conversely, $p = 3$ works with $n = 1$, since $1 + 1 + 1 = 3$. If $p \equiv 1 \pmod 9$, the Primitive Root Theorem gives an n of order 9 modulo p. Then $p \mid n^9 - 1 = (n^3 - 1)(n^6 + n^3 + 1)$, while $p \nmid n^3 - 1$ because the order is not a divisor of 3. Hence $p \mid n^6 + n^3 + 1$.

So the primes are 3 together with the primes below 200 that are 1 modulo 9. Running through $10, 19, 28, \ldots, 199$, these are $19, 37, 73, 109, 127, 163, 181, 199$ (the others, such as $91 = 7 \cdot 13$ and $145 = 5 \cdot 29$, are composite or even). The sum is $3 + 19 + 37 + 73 + 109 + 127 + 163 + 181 + 199 = 911$.

Answer: 911

Problem 5

Prove that there are infinitely many primes of the form $8k + 1$.

Suppose, for contradiction, that there are only finitely many primes congruent to 1 modulo 8, and call them $p_1, \ldots, p_r$ (possibly $r = 0$). Let $N = 2p_1p_2 \cdots p_r$, with $N = 2$ if $r = 0$, and let p be any prime factor of $N^4 + 1$.

Since N is even, $N^4 + 1$ is odd, so p is odd. Also $p \nmid N$, since otherwise $p \mid 1$. From $N^4 \equiv -1 \pmod p$ we get $N^8 \equiv 1$, so $d = \text{ord}_p(N)$ divides 8. If d divided 4 we would have $N^4 \equiv 1$, hence $1 \equiv -1$ and $p \mid 2$, which is false. Therefore $d = 8$. By Fermat's Little Theorem and the Order Lemma, $8 \mid p - 1$, so $p \equiv 1 \pmod 8$.

So p is one of $p_1, \ldots, p_r$, and hence $p \mid N$. This contradicts $p \nmid N$. Therefore there are infinitely many primes of the form $8k + 1$. □

Problem 6

Let n be an odd positive integer such that n divides $3^n + 1$. Prove that $n = 1$.

Suppose, for contradiction, that $n > 1$, and let p be the smallest prime factor of n. Since n is odd, p is odd. Also $p \neq 3$, because $3^n + 1 \equiv 1 \pmod 3$ and p divides $3^n + 1$.

From $p \mid 3^n + 1$ we get $3^n \equiv -1$ and so $3^{2n} \equiv 1 \pmod p$. Let $d = \text{ord}_p(3)$, which exists as $p \neq 3$. By the Order Lemma $d \mid 2n$, and by Fermat's Little Theorem $d \mid p - 1$. Hence $d \mid \gcd(2n, p - 1)$.

We claim $\gcd(2n, p - 1) = 2$. Let q be an odd prime dividing this gcd. Then $q \mid n$, so $q \geq p$ by the

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choice of p, but also $q \mid p-1$, so $q \leq p-1$. This is impossible, so the gcd is a power of 2. As n is odd, $4 \nmid 2n$, so the gcd divides 2, and the gcd equals 2 since $p-1$ is even.

Therefore $d \mid 2$, so $3^2 \equiv 1 \pmod{p}$, that is, $p \mid 8$. This contradicts p being odd. Hence $n = 1$, which does satisfy $1 \mid 3+1$. $\square$

Remark. The oddness of n is needed, since $n = 2$ and $n = 10$ both divide $3^n + 1$, since $3^{10} + 1 = 59050$.

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